

Bitwise differential trail through **QALQAN** found by SAT

CiVerLy

August 4, 2026

1 QALQAN

Maximal differential probability: 2^{-74}

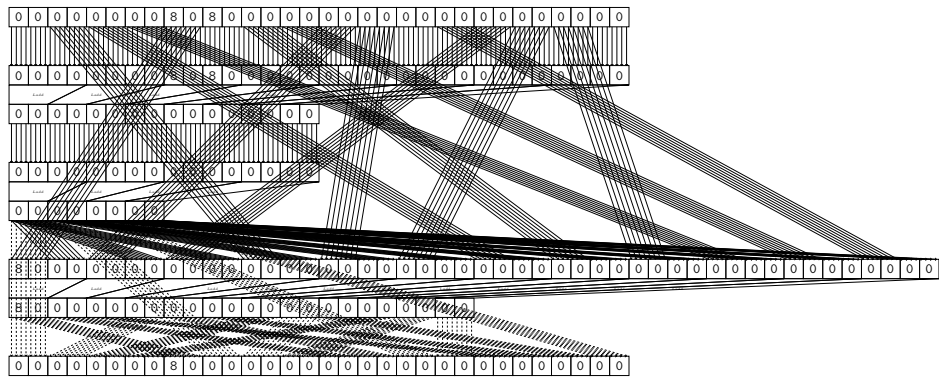
0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
K register																												
0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S register																												
0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
T register																												
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
U																												
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
V																												
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
X register																												
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Y register																												
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Z register																												
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A register																												
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B register																												
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C register																												
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0
D																												
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0

2 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

3 L

Maximal differential probability: $2^{-0.0}$



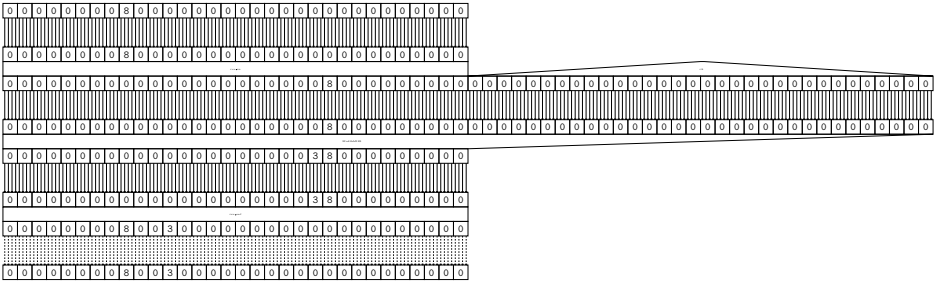
4 QalqanRound

Maximal differential probability: $2^{-19.0}$

0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

5 Add128

Maximal differential probability: $2^{-3.0}$



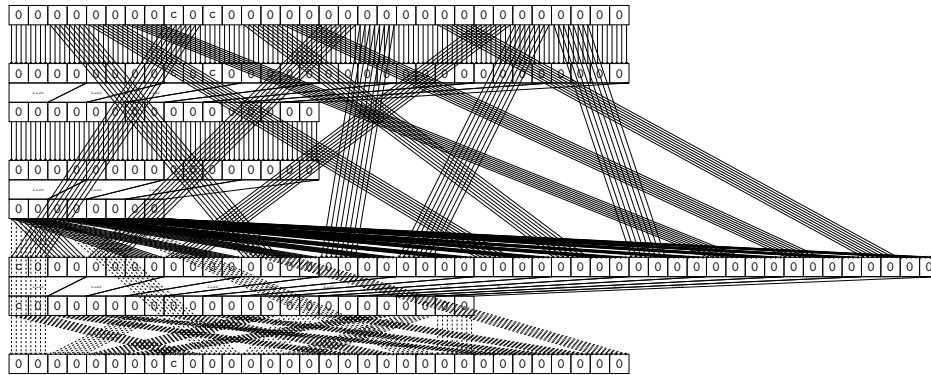
6 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	c	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7 L

Maximal differential probability: $2^{-2.0}$



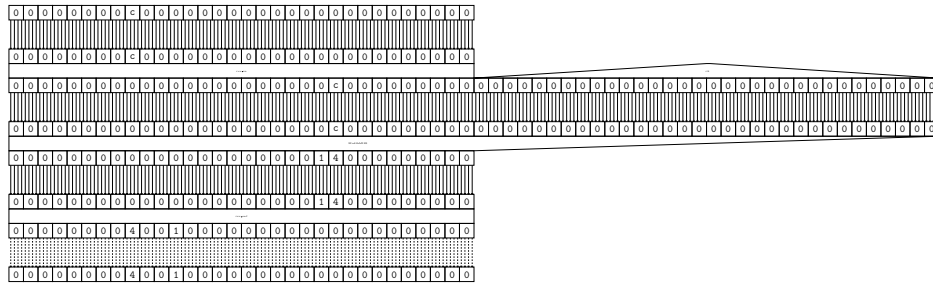
8 QalqanRound

Maximal differential probability: $2^{-17.0}$

0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AES128																													
0	0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2RoundLayer																													
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L																													
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

9 Add128

Maximal differential probability: $2^{-3.0}$

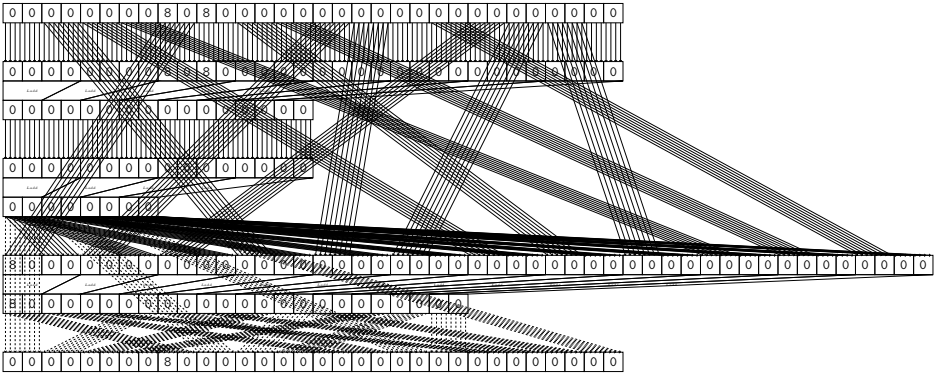


10 SBoxLayer

Maximal differential probability: $2^{-14.0}$ [illegible]

11 L

Maximal differential probability: $2^{-0.0}$



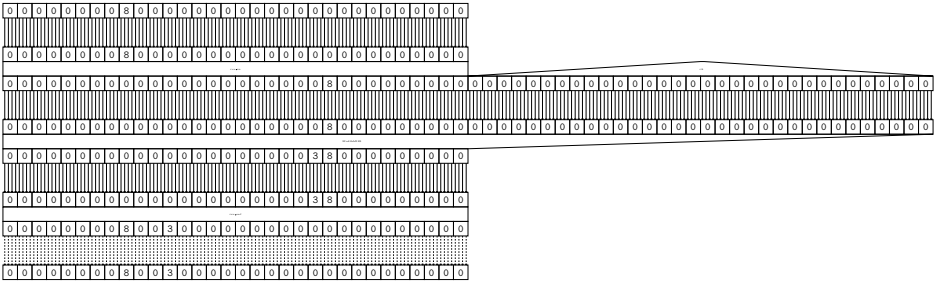
12 QalqanRound

Maximal differential probability: $2^{-18.0}$

0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AES128																												
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2RoundKey																												
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L																												
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13 Add128

Maximal differential probability: $2^{-3.0}$



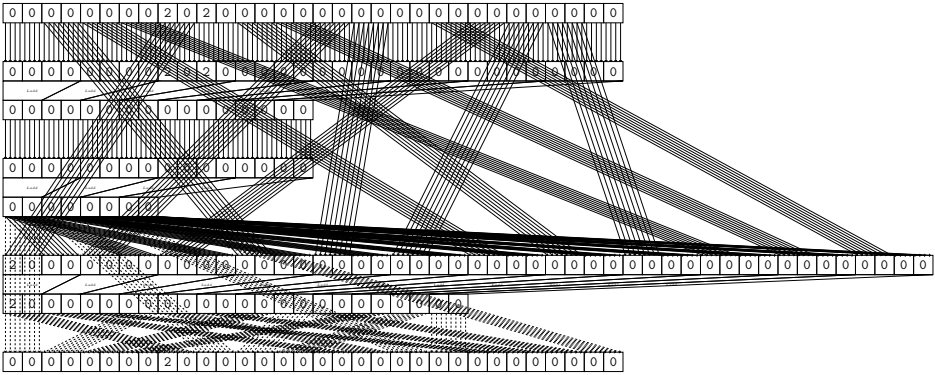
14 SBoxLayer

Maximal differential probability: $2^{-13.0}$

0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15 L

Maximal differential probability: $2^{-2.0}$



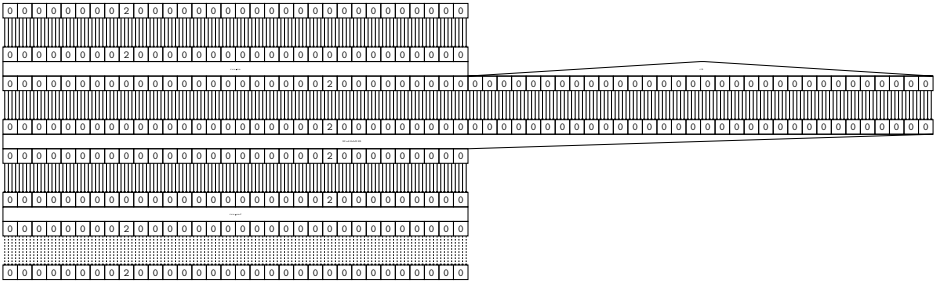
16 QalqanRound

Maximal differential probability: $2^{-8.0}$

0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
AES128																												
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
2RoundKey																												
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
L																												
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	8	0	0	0	0	

17 Add128

Maximal differential probability: $2^{-1.0}$



18 SBoxLayer

Maximal differential probability: $2^{-7.0}$

0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

19 L

Maximal differential probability: $2^{-0.0}$

